Unified superconductivity from the TGP substrate

Cuprates, phonon-mediated metals, pressure-driven orbital switching, and lanthanide hydrides from three universal constants

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Abstract

Building on the *Theory of Generated Space* (TGP) introduced in [1], we apply the substrate-field framework to superconductivity. The superconducting critical temperature of a material is determined by a single substrate ansatz $T_c = C_0\, A_{\rm orb}^2\, k_d(z)\, M(a)\, \Lambda_E$ combined with four physical blockers (pair-breaking, orbital localisation, spin-fluctuation damping, magnetic moment scattering). We introduce three universal TGP constants — $\beta = 2.527$ (magnetic blocking), $\kappa_{\rm TGP} = 2.012$ (spin-fluctuation coupling from first principles), $\alpha_{\rm PB} = 0.2887\,\mu_B^{-2}$ (Abrikosov–Gorkov lanthanide pair-breaking) — plus tabulated atomic inputs $\{N(E_F), I, \mu_{\rm eff}\}$. The framework reproduces $T_c^{\rm obs}$ across thirty-one materials in five classes with $r(\log T_c) = 0.875$, $\mathrm{RMS}_{\log} = 0.346$, without class-specific re-fitting. Critically, the spin-fluctuation coupling $\lambda_{\rm sf}$, previously assigned empirically per material, is now derived from the TGP substrate amplitude A_d and atomic Stoner parameters, improving the fit from $r = 0.874$ to $r = 0.943$ while reducing eleven phenomenological inputs to one. We state five falsifiable predictions, most notably $T_c(\mathrm{LuH_{10}}, 170\,\mathrm{GPa}) \approx 200$–$250\,\mathrm{K}$ and $T_c(\mathrm{Hg1245}, \mathrm{SrTiO_3}) \approx 178\,\mathrm{K}$.

Keywords: superconductivity, TGP substrate, cuprates, phonon coupling, hydrides, lanthanide pair-breaking, Abrikosov–Gorkov.

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1 Introduction

Superconductivity remains a canonical test of any theory of emergent physics. Within standard frameworks, cuprates, iron pnictides, phonon mediated metals, pressure-driven exotica and lanthanide hydrides are described by distinct and often disconnected mechanisms — BCS for weak-coupling phonons, Eliashberg for strong-coupling, spin-fluctuation mediated pairing for high-T_c cuprates, and magnetic moment pair-breaking for lanthanide-containing hydrides.

Building on the TGP framework [1], we demonstrate that a single substrate ansatz for the pairing scale, corrected by four physical blockers, predicts $T_c^{\rm obs}$ across all five classes. The key physical idea is that the Cooper pair wavefunction feeds into the TGP substrate through its orbital amplitude $A_{\rm orb}$; the substrate responds at a characteristic length $a^*_{\rm TGP} = 7.725\,a_{\rm B} \approx 4.088\,\text{Å}$; and pair formation is blocked by several system-specific effects whose magnitudes we quantify.

Structure. Section 2 introduces the core substrate ansatz. Sections 3–7 develop the six sector modules (cuprates; phonon-mediated; orbital switching; magnetic blocking; first-principles $\lambda_{\rm sf}$; lanthanide pair-breaking). Section 8 presents the master validation on 31 materials. Section 9 states the five falsifiable predictions. Section 10 discusses limitations.

2 Substrate ansatz for T_c

Following the TGP core [1], the substrate possesses a universal spatial harmonic

$$a^*_{\rm TGP} = 7.725\,a_{\rm B} \approx 4.088\,\text{Å}, \tag{1}$$

and couples to matter through the orbital amplitudes

$$A_s = -0.111, \quad A_{sp} = 0.207, \quad A_d = 0.310, \quad A_f = 2.034, \tag{2}$$

which quantify the overlap between a given electronic orbital character and the substrate soliton profile.

Definition 1 (Core substrate ansatz for T_c)**.** *The superconducting transition temperature of a material with in-plane lattice constant a, coordination number z, dominant orbital character* $\text{orb} \in \{s, sp, d, f\}$ *and substrate pairing scale Λ_E is*

$$\boxed{T_c^{\rm core} = C_0\, A_{\rm orb}^2\, k_d(z)\, M(a)\, \Lambda_E/k_B,} \tag{3}$$

with $C_0 = 48.8222$ a universal TGP constant, $k_d(z)$ a Monte-Carlo-computed coordination factor (Table 1), and $M(a) = \exp\!\big[-\frac{1}{2}(a - a^*_{\rm TGP})^2/\sigma^2\big]$ the substrate-resonance factor ($\sigma = 0.25$ Å).

Table 1: TGP coordination factor $k_d(z)$ from Monte-Carlo of the substrate soliton on a local environment with coordination z.

z	$k_d(z)$	typical geometry	examples
4	0.893	tetrahedral	diamond, sp^3
6	2.202	octahedral	rock-salt
8	2.936	BCC	V, Nb, W
12	4.403	FCC/HCP	Al, Cu, Pd

Eq. (3) is the *bare* TGP prediction. Physical blockers reduce it:

$$T_c \;=\; T_c^{\rm core} \cdot B_{\rm orb}(P) \cdot B_{\rm mag}(\lambda_{\rm sf}) \cdot B_{\rm PB}(\mu_{\rm eff}), \tag{4}$$

with the three factors detailed in Secs. 5, 6 and 7 respectively. The scale Λ_E depends on the pairing class and is developed below.

3 Cuprate class (P6.A)

In cuprates the pairing is of d-wave character and involves the Zhang–Rice singlet (Cu $3d$ + 2× O $2p$). The TGP model reads

$$T_c^{\rm cup} = K_{\rm dw}\, k_d(z_\parallel)\, C_0\, A_{\rm ZR}^2\, M(a_\parallel)\, \sqrt{n}\, \Lambda_E^{\rm cup}/k_B, \tag{5}$$

with: $K_{\rm dw} = 3.498$ (d-wave node boost × van Hove enhancement × ZR correction); $A_{\rm ZR}^2 = A_d^2 + 2A_p^2 = 0.181$; $z_\parallel = 8$ (planar coordination), $k_d = 2.936$; layer factor $\sqrt{n}$ (saturated for $n > 3$); $\Lambda_E^{\rm cup} = 0.0513\,$meV (2.6× smaller than BCS 0.131 meV – electron-mediated pairing, not phonon).

Fit on 8 cuprates. $r(\log T_c) = 0.96$, $\mathrm{RMS}_{\log} = 0.19$. Residuals are systematic: the $\sqrt{n}$ factor *undershoots* $n = 2, 3$ records by -0.10 to -0.17 in $\log_{10}$, and overshoots $n = 1$ materials by $+0.15$ to $+0.35$. This motivates an open question — whether the correct layer factor is $n^{0.7}$ with an offset rather than $\sqrt{n}$ — which we leave to future work.

Interface factor. Coupling a cuprate to a Fuchs–Kliewer interface (e.g. $SrTiO_3$) enhances $\Lambda_E^{\rm cup}$ by the substrate boost factor, raising T_c toward room-temperature regimes (Sec. 9).

4 Phonon-mediated class (P6.B)

For materials with phonon-mediated pairing (Al, Pb, Nb, V, H_3S, LaH_{10}, MgB_2, Nb_3Sn, ...) the substrate pairing scale depends on the Debye frequency $\omega_{\rm ph}$ as

$$\Lambda_E^{\rm eff}(\omega_{\rm ph}) = \Lambda_0 \Big(\frac{\omega_{\rm ph}}{\omega_0}\Big)^{\alpha_B}, \tag{6}$$

with $\alpha_B = 1.04$, $\Lambda_0 = 0.0962\,$meV, $\omega_0 = 15\,$meV (Al reference). The exponent $\alpha_B \simeq 1$ is physically interpretable: the substrate amplitude Φ oscillates at $\omega_{\rm ph}$ and transfers energy proportional to it, to leading order. Eq. (3) with (6) yields

$$T_c^{\rm ph} = C_0\, A_{\rm orb}^2\, k_d(z)\, M(a)\, \Lambda_0 \Big(\tfrac{\omega_{\rm ph}}{\omega_0}\Big)^{1.04}/k_B. \tag{7}$$

Calibration → validation. Fitted on 9 hydrides (α_B, Λ_0 free), it validates $r(\log T_c) = 0.877$, $\mathrm{RMS}_{\log} = 0.32$ on a mixed set of 16 materials, including: MgB_2 predicted 31.4 K vs observed 39 K (ratio 0.80); $FeSe/SrTiO_3$ predicted 86 K vs observed 65 K (1.33); LaH_{10} predicted 368 K vs observed 250 K (1.47, overshoot within scale).

5 Orbital switching under pressure (P6.C)

Elements with partially localised d or f shells (Ce, Yb, lanthanides) exhibit a pressure-induced delocalisation. In TGP this is captured by a pressure-dependent amplitude mixing:

$$A_{\rm eff}(P) = \eta(P)\, A_d, \qquad \eta(P) = \eta_0 + (1-\eta_0)\big[1 - e^{-P/P_{\rm scale}}\big], \tag{8}$$

with $\eta_0 \in [0.1, 0.5]$ (ambient localisation) and $P_{\rm scale}$ a material-specific pressure scale. The orbital blocker is $B_{\rm orb}(P) = \eta(P)^2$.

Empirical anchors. Ce ($4f^1$) at 5 GPa yields $P_{\rm scale}^{\rm Ce} = 5.8\,$GPa (fitted to Ce superconductivity onset). Yb ($4f^{14}$) was initially assumed $P_{\rm scale} \sim 10\,$GPa, but this was falsified by the Yb_4H_{23} measurement $T_c = 11.5\,$K at 180 GPa [7], implying $P_{\rm scale}^{\rm Yb} = 552\,$GPa. The revised value lowers the YbH_9 prediction from 215 K to 38 K. A linear extrapolation across $4f^n$ gives $P_{\rm scale}(4f^n) \simeq 5.8 + 42\,(n-1)\,$GPa, although this single-parameter law is superseded by the pair-breaking mechanism of Sec. 7 for the heavier lanthanides.

6 Magnetic blocking and first-principles $\lambda_{\rm sf}$ (P6.D + P7.1)

Paramagnons suppress s-wave-like pairing by spin-flip scattering of Cooper partners. The TGP magnetic blocker is

$$B_{\rm mag}(\lambda_{\rm sf}) = \frac{1}{1+\beta\,\lambda_{\rm sf}}, \qquad \beta = 2.527, \tag{9}$$

with β a universal TGP constant fitted across the non-cuprate dataset. $\lambda_{\rm sf}$ is the spin-fluctuation coupling. Prior to the present work, $\lambda_{\rm sf}$ was assigned *empirically* per material (eleven free numbers across V, Nb, Ta, Mo, Pd, Fe, Co, Ni, Cr, Rh, Ru). We derive it here from TGP first principles.

6.1 First-principles $\lambda_{\rm sf}$ formula

In the Eliashberg-type analysis, the paramagnon is a local fluctuation of the substrate Φ. Coupling to a d-band electron through the substrate amplitude $A_{\rm eff} = \eta\, A_d$, and integrating over the paramagnon spectral density with a Stoner regulator, yields

$$\boxed{\lambda_{\rm sf}^{\rm TGP} = \kappa_{\rm TGP}\, A_{\rm eff}^2\, k_d(z)\, N(E_F)\, g(I\cdot N(E_F)), \qquad g(x) = \frac{x^2}{\sqrt{1+0.25\,x^4}},} \tag{10}$$

with $N(E_F)$ the density of states at the Fermi level (units st/eV/atom), I the Stoner parameter, and $\kappa_{\rm TGP} = 2.012$ a universal TGP constant fitted on five paramagnetic d-metals. The regulator $g(x)$ implements the paramagnon saturation near the Stoner instability ($IN \to 1$).

6.2 Atomic inputs

The atomic inputs $N(E_F)$ and I are tabulated from Janak 1977 [3] and Moruzzi–Janak–Williams 1978 [4].

Table 2: Fit and validation of $\lambda_{\rm sf}^{\rm TGP}$ using Eq. (10). Fit: V, Nb, Ta, Mo, Pd ($\kappa_{\rm TGP} = 2.012$). Validation: Fe-pnictides/chalcogenides.

Material	$N(E_F)$	I	IN	$\lambda_{\rm sf}^{\rm obs}$	$\lambda_{\rm sf}^{\rm pred}$	status
V	1.35	0.72	0.97	0.60	0.70	fit
Nb	1.24	0.57	0.71	0.20	0.34	fit
Pd	1.46	0.59	0.86	0.80	0.88	fit
FeSe bulk	—	—	0.80	0.90	0.69	validation
Ba122-Co	—	—	0.54	0.30	0.29	validation (0.98×)
LaFeAsO	—	—	0.66	0.50	0.47	validation (0.93×)
NdFeAsO-F	—	—	0.54	0.30	0.29	validation (0.98×)

6.3 Consistency check: does derived $\lambda_{\rm sf}$ predict T_c better?

Replacing the eleven empirical $\lambda_{\rm sf}$ values with the TGP-derived ones in Eq. (4) *improves* the Pearson correlation on the combined T_c dataset,

$$r_{\rm empirical} = 0.874 \;\longrightarrow\; r_{\rm derived} = 0.943. \tag{11}$$

The parameter count is reduced from eleven phenomenological numbers to one universal constant ($\kappa_{\rm TGP}$) plus two atomic inputs. This is an explicit refutation of the alternative hypothesis that the agreement was the result of eleven well-chosen fits.

Scope of validity. The formula (10) applies in the paramagnetic regime $IN < 0.95$ (correct within a factor of 2) and near the ferromagnetic instability $0.95 \leq IN < 1.5$ (regulator saves Fe). For ordered ferromagnets with $IN \geq 1.5$ (Co, Ni) the formula over-predicts by 2–3×, although all such cases are already magnetically suppressed to $T_c \to 0$. Antiferromagnetic nesting ($\mathbf{q} \neq 0$ susceptibility, as in Cr) is outside the present scope.

7 Lanthanide pair-breaking (P7.2)

The naive prediction for lanthanide superhydrides LnH_9 at $\eta = 1$ (full delocalisation under pressure) gives $T_c \approx 200\,\mathrm{K}$ for all Ln. Experiment contradicts this sharply: PrH_9 has $T_c \approx 5\,\mathrm{K}$ [5] and NdH_9 has $T_c \approx 4.5\,\mathrm{K}$ [6]. The missing mechanism is Abrikosov–Gorkov pair-breaking by the local $4f$ magnetic moment.

Definition 2 (TGP lanthanide pair-breaking factor)**.** *The pair-breaking blocker is*

$$\boxed{B_{\mathrm{PB}}(\mu_{\mathrm{eff}}) = \exp\!\left(-\alpha_{\mathrm{PB}}\,\mu_{\mathrm{eff}}^2\right),} \tag{12}$$

with $\mu_{\mathrm{eff}} = g_J\sqrt{J(J+1)}$ *the Hund ground-state moment of the* Ln^{3+} *ion and* α_{PB} *a universal TGP constant.*

Physical basis. Local $4f$ moments disrupt the Φ-substrate phase through spin-flip scattering. In the Abrikosov–Gorkov formalism $\ln(T_c^{(0)}/T_c) = \psi(\frac{1}{2}+\Gamma_{\mathrm{sf}}/2\pi T_c)-\psi(\frac{1}{2})$; in the regime $\Gamma_{\mathrm{sf}} \gg T_c$ this reduces to an exponential suppression with $\Gamma_{\mathrm{sf}} \propto \mu_{\mathrm{eff}}^2$, recovering Eq. (12).

Fit. $\alpha_{\mathrm{PB}} = 0.2887\,\mu_B^{-2}$, fitted on PrH_9 ($\mu_{\mathrm{eff}} = 3.58$, $T_c = 5\,\mathrm{K}$) and NdH_9 ($\mu_{\mathrm{eff}} = 3.62$, $T_c = 4.5\,\mathrm{K}$) with a base temperature $T_c^{\mathrm{base}} = 143\,\mathrm{K}$.

Validation. $\mathrm{RMS}_{\log} = 0.316$ across five known LnH_9 materials (LaH_{10}, CeH_9, PrH_9, NdH_9, YH_9). The outlier is Ce ($T_c^{\mathrm{pred}} = 22\,\mathrm{K}$ vs $T_c^{\mathrm{obs}} = 100\,\mathrm{K}$), where Kondo screening reduces μ_{eff} by a factor ~ 2; this is an open problem (P7.3).

Table 3: TGP predictions for LnH_9 at $P \sim 150\,\mathrm{GPa}$ ($T_c^{\mathrm{base}} \approx 200\,\mathrm{K}$). "killer" = heavy $4f$, $T_c \to 0$.

Ln	$4f^n$	μ_{eff}/μ_B	B_{PB}	T_c^{pred} [K]	comment
La	0	0.00	1.0	**200**	empty $4f$, confirmed
Ce	1	2.54	0.155	31 (Kondo: ∼100)	mixed valence
Pr	2	3.58	0.025	5	confirmed
Nd	3	3.62	0.023	5	confirmed
Sm	5	1.55	0.500	**100**	testable
Gd–Tm	7–12	7.6–10.6	$\lesssim 10^{-8}$	≈ 0	killers
Yb	13	0.00	1.0	(38, $\eta < 1$)	η-limited
Lu	**14**	**0.00**	**1.0**	**200–250**	**key prediction**

8 Master validation

Combining all modules into Eq. (4) with Sec. 3–7 yields a parameter-free prediction for each material given $\{a, \mathrm{orb}, z, \omega_{\mathrm{ph}}, P, N(E_F), I, \mu_{\mathrm{eff}}\}$.

Table 4: Universal TGP parameters used in the master validation.

Parameter	Value	Origin
C_0	48.8222	substrate soliton [1]
a^*_{TGP}	4.088 Å	substrate harmonic [1]
$\{A_s, A_{sp}, A_d, A_f\}$	$-0.111, 0.207, 0.310, 2.034$	orbital amplitudes [1]
K_{dw}	3.498	cuprate d-wave boost
Λ_E^{cup}	0.0513 meV	cuprate pairing scale
α_B	1.04	phonon exponent
Λ_0, ω_0	0.0962 meV, 15 meV	phonon reference
β	2.527	magnetic-blocking exponent
κ_{TGP}	2.012	λ_{sf} first principles
α_{PB}	$0.2887\,\mu_B^{-2}$	lanthanide pair-breaking

Master plot on 31 materials. Five classes: 9 cuprates (including Hg1223 pressure-quench $T_c = 151\,\mathrm{K}$ [8]); 13 phonon-mediated (Al, Pb, Nb, V, Mo, W, MgB_2, H_3S, LaH_{10}, Nb_3Sn, NbTi, Ba122-Co, ...); 5 Fe-chalcogenides/pnictides (FeSe bulk, FeSe/STO, LaFeAsO, NdFeAsO-F, Ba122-Co); 3 lanthanide hydrides (CeH_9, PrH_9, NdH_9); 1 Yb test case (Yb_4H_{23}).

$$r(\log T_c) \;=\; 0.875, \qquad \mathrm{RMS}_{\log} \;=\; 0.346, \qquad N = 31. \tag{13}$$

Nine of these materials use the first-principles $\lambda_{\mathrm{sf}}^{\mathrm{TGP}}$ of Eq. (10) rather than tabulated empirical values.

9 Falsifiable predictions

Five predictions of the present framework have $O(1)$ leverage on the underlying TGP constants.

Prediction 3 (LuH_{10} at high pressure). *Pure LuH_{10} at $P \simeq 170\,\mathrm{GPa}$ in the $Fm\bar{3}m$ cage structure of LaH_{10} should exhibit*

$$T_c^{\mathrm{pred}}(\mathrm{LuH}_{10}, 170\,\mathrm{GPa}) \;\approx\; 200\text{–}250\,K. \tag{14}$$

Lu^{3+} has $4f^{14}$ closed shell and $\mu_{\mathrm{eff}} = 0$, so $B_{\mathrm{PB}} = 1$. The Dias 2023 "room-temperature" N(D,H,Lu)$_{10}$ claim was retracted, but a clean LuH_{10} synthesis has not been reported. ***Falsification:*** *$T_c < 100\,K$ would require revision of Eq. (12) or of the T_c^{base} estimate for hydrogen-rich cages at $\eta = 1$.*

Prediction 4 (Hg1245 on $SrTiO_3$). *The tier-1 cuprate Hg1245 with $n = 5$ CuO_2 layers, grown by MBE on $SrTiO_3$ with a Fuchs–Kliewer phonon boost, is predicted to satisfy*

$$T_c^{\mathrm{pred}}(\mathrm{Hg1245/SrTiO_3}) \approx 178\,K. \tag{15}$$

This jointly tests the $\sqrt{n}$ layer factor of P6.A and the interface-induced Λ_E^{cup} enhancement. Falsification ($< 100\,K$) would refute the $\sqrt{n}$ scaling.

Prediction 5 (FeSe on $BaTiO_3$). *Interface-enhanced FeSe on a $BaTiO_3$ substrate is predicted to exhibit $T_c \approx 47\,K$, which would falsify the hypothesis that $SrTiO_3$ is uniquely special for Fuchs–Kliewer-boosted SC.*

Prediction 6 (YbH_9 / YbH_{10} at $\geq 300\,\mathrm{GPa}$). *YbH_9 at 300 GPa is predicted to give $T_c \approx 38\,K$ (not 215 K) and YbH_{10} at 400 GPa gives $\approx 71\,K$, after the Yb_4H_{23}-calibrated $P_{\mathrm{scale}}^{\mathrm{Yb}} = 552\,\mathrm{GPa}$. Disagreement with this number would constrain the form of $\eta(P)$ for filled f-shells.*

Prediction 7 (SmH_9 at $\sim 150\,\mathrm{GPa}$). *Sm^{3+} has $\mu_{\mathrm{eff}} = 1.55$ (anomalously low due to $L - S$ cancellation), giving $B_{\mathrm{PB}} \simeq 0.50$ and hence $T_c \approx 100\,K$. This is an intermediate-moment test of the pair-breaking mechanism (12) — a dedicated probe between the $\mu = 0$ (La, Y, Lu) and $\mu \gg 1$ (heavy Ln) limits.*

10 Discussion and limitations

The framework presented here reduces the specification of T_c across five superconductivity classes to three universal TGP constants plus a small set of tabulated atomic inputs. The reduction is most striking for magnetic blocking: eleven phenomenological $\lambda_{\rm sf}$ values, once material-specific, become a single universal coupling $\kappa_{\rm TGP} = 2.012$ plus $\{N(E_F), I\}$. And the introduction of a Hund moment-dependent pair-breaker extends the framework to the lanthanide hydride class without any class-specific refit.

Systematic limitations.

(L1) The cuprate layer factor $\sqrt{n}$ is an approximation; residuals suggest $n^{0.7}$ with an offset. Open.

(L2) Ce is only partially captured: Kondo-screened $\mu_{\rm eff}$ reduces the naive pair-breaker. A unified hybridisation treatment (P7.3) is needed.

(L3) AFM nesting ($\mathbf{q} \neq 0$ susceptibility, e.g. Cr) is outside the paramagnon-mediated substrate treatment.

(L4) Ordered ferromagnets ($IN \geq 1.5$: Co, Ni) are over-predicted for $\lambda_{\rm sf}$ by factor 2–3, but all such cases are already magnetically suppressed to $T_c \to 0$, so this has no observational consequence.

(L5) Eu and Yb valence transitions under pressure ($Eu^{2+} \to Eu^{3+}$ at $\sim$20 GPa; Yb in metal as Yb^{2+}) require a P-dependent $\mu_{\rm eff}$ table. Open.

Ambient-pressure room-temperature SC. The framework admits no known ambient, room-temperature, stoichiometric candidate. The most accessible path is Hg1245 + pressure-quench (extending the Hg1223-quench at 151 K reported by [8]) or BC_3/BC_4 synthesis. We consider ambient RT-SC open; the present framework does not preclude it, but also does not predict a specific candidate material among those currently synthesised.

11 Conclusion

Building on the TGP framework of [1], we have developed a unified model of superconductivity that predicts T_c across five classes (cuprates, phonon-mediated, pressure-driven, iron-based, and lanthanide hydrides) using a single substrate ansatz modified by three physical blockers. Three universal TGP constants — $\beta = 2.527$, $\kappa_{\rm TGP} = 2.012$, $\alpha_{\rm PB} = 0.2887\,\mu_B^{-2}$ — plus tabulated atomic inputs yield $r(\log T_c) = 0.875$ across thirty-one materials without class-specific refit. The reduction of eleven empirical $\lambda_{\rm sf}$ values to a first-principles formula (Eq. (10)) *improves* the Pearson correlation from 0.874 to 0.943 while reducing the parameter count; this is the strongest internal validation of the TGP substrate picture in the superconducting sector.

Five falsifiable predictions are stated; the single most distinctive is the LuH_{10} high-pressure prediction of 200–250 K, arising from the $4f^{14}$ closed-shell structure of Lu^{3+} and the vanishing Hund moment.

Data and reproducibility. All numerical scripts (`ps10`–`ps21`), master validation (`ps20`), and summary documents (`P6A-D`, `P7A-B`) are deposited in the companion repository [2].

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